

Robot Arm & Claw · Premium Course

2 live sessions · 2 hrs each · 12-15 yrs · from KES 7,000

Premium course outline **PREMIUM**

Overview

Assemble and program a 3-DOF robot arm with a gripper claw and a joystick controller.

DETAIL

Track	Arduino Robotics & Electronics
Format	2 live sessions · 2 hrs each
Delivery	Live online (1:1 or small group) or in person, you choose when booking
Ages	12-15 yrs
Level	Advanced
Price	from KES 7,000
Maker Kit	Optional, KES 8,500 one-time: Arduino Uno R3 board + USB cable, Breadboard + 40 jumper wires, LEDs (red, yellow, green, white, RGB) + resistors, Push buttons, potentiometer and piezo buzzer, Light sensor (LDR) and sound sensor, HC-SR04 ultrasonic sensor + bracket, IR sensor, DHT11 temperature & humidity sensor, PIR motion sensor, SG90 servo motor, Relay module, 16x2 LCD screen (I2C), Battery holder, Storage box. Bought once and reused in every Arduino course. Needed at home for live online sessions; for in-person sessions we provide kits, so buy one only to keep building at home.
Parts pack	Optional, KES 2,500: 3-DOF robot arm frame + claw, 4 extra SG90 servos, 2 joystick modules
Standalone	No
Prerequisites	Servo motor control (free)

Programme

- 1. Build & Wire.** Assemble a 3-DOF (degree-of-freedom) arm using SG90 servos and build a joystick controller with two analog sticks.

2. **Program & Challenge.** Calibrate movement ranges and add a gripper claw servo. Challenge: pick up and sort coloured objects by zone.

What they take home

 Assembled robot arm

Why parents love it

“The same principles behind factory robots and surgical arms — in your child's hands.”

How to book

Places are limited and run on scheduled dates. To book or ask a question, email **tinkerwithanne@gmail.com** with the course name (**Robot Arm & Claw**), your child's age, and your preferred dates. Prices start from the amount above and may vary with group size and materials.